

Mon, Jan 19

Properties

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x) \cdot g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x) \neq 0}$$

Composite Functions

$$(f \circ g)(x) = f(g(x))$$

$$(g \circ f)(x) = g(f(x))$$

$$2) \quad f(x) = 2 + 3x - x^2$$

$$g(x) = 2x - 1$$

$$1) \quad (f+g)(4)$$

$$f(x) = 2 + 3x - x^2$$

$$g(x) = 2x - 1$$

$$2 + 3x - x^2 + 2x - 1$$

$$-x^2 + 5x + 1$$

$$-(4)^2 + 5(4) + 1$$

$$= 5$$